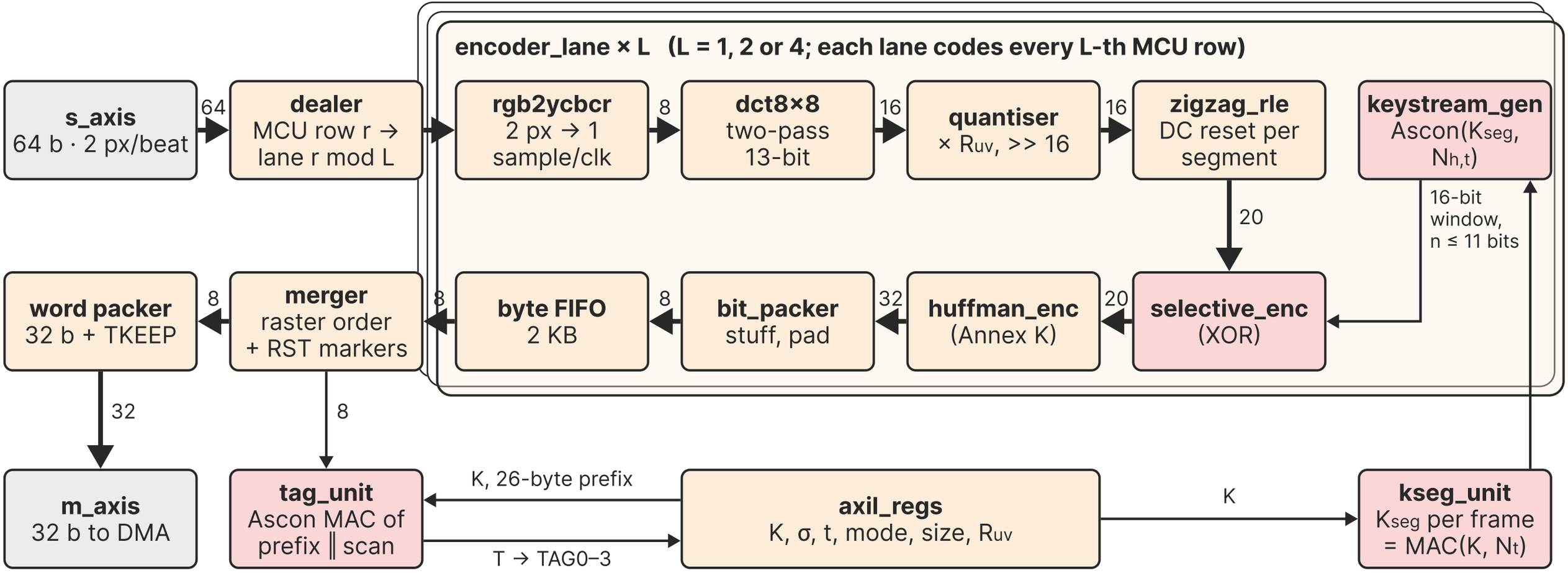

encoder_lane × L (L = 1, 2 or 4; each lane codes every L-th MCU row)
s_axis
64 b · 2 px/beat
64
dealer
MCU row r → lane r mod L
rgb2ycbcr
2 px → 1 sample/clk
8
dct8×8
two-pass 13-bit
16
quantiser
× Ruv, >> 16
16
zigzag_rle
DC reset per segment
keystream_gen
Ascon(Kseg, Nh,t)
20
16-bit window, n ≤ 11 bits
word packer
32 b + TKEEP
8
merger
raster order + RST markers
8
byte FIFO
2 KB
8
bit_packer
stuff, pad
32
huffman_enc
(Annex K)
20
selective_enc
(XOR)
32
8
m_axis
32 b to DMA
tag_unit
Ascon MAC of prefix ‖ scan
K, 26-byte prefix
T → TAG0–3
axil_regs
K, σ, t, mode, size, Ruv
K
kseg_unit
Kseg per frame = MAC(K, Nt)